

The effect of supplemental vitamins and minerals on the development of prostate cancer: a systematic review and meta-analysis.



BACKGROUND: Vitamin supplementation is used for many purposes with mainly alleged benefits. One of these is the use of various vitamins for the prevention of prostate cancer. METHODS: We conducted a systematic review and meta-analysis on this topic. Pubmed, Embase and the Cochrane Database were searched; as well, we hand searched the references in key articles. Randomized controlled trials (RCTs), cohort studies and case-control studies were included. The review assessed the effect of supplemental vitamins on the risk of prostate cancer and on disease severity and death in men with prostate cancer. RESULTS: Fourteen articles were included in the final assessment. Individually, a few of these studies showed a relationship between the ingestion of supplemental vitamins or minerals and the incidence or severity of prostate cancer, especially in smokers. However, neither the use of multivitamin supplementation nor the use of individual vitamin/mineral supplementation affected the overall occurrence of prostate cancer or the occurrence of advanced/metastatic prostate cancer or death from prostate cancer when the results of the studies were combined in a meta-analysis. We also conducted several sensitivity analyses by running meta-analysis using just the higher quality studies and just the RCTs. There were still no associations found. CONCLUSIONS: There is no convincing evidence that the use of supplemental multivitamins or any specific vitamin affects the occurrence or severity of prostate cancer. There was high heterogeneity among the studies so it is possible that unidentified subgroups may benefit or be harmed by the use of vitamins.